Mahmoud Mohamed Bassam Alsyeed Alsalahy Ahmed

Contact Information: - **Address:** Giza, Egypt - **Mobile:** +20-1277939786 - **Email:** mahmoudbassam1999@gmail.com - **LinkedIn:** [LinkedIn Profile] - **GitHub:** [GitHub Profile] - **Nationality:** Egyptian - **Military Status:** Exempted

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Objective

AI-oriented Mechatronics Engineer with solid hands-on experience in machine learning, embedded systems, and digital twin technologies. Currently building an AI-powered assistant using RAG architecture for CNC maintenance and customer support at Threecuts Company. Passionate about developing data-driven solutions for industrial automation and real-time diagnostics.

Professional Experience

AI & Maintenance Engineer
Threecuts Company, Cairo, Egypt
Jul 2025 – Present - Leading the development of an intelligent CNC assistant using RAG to support machine maintenance and customer queries. - Training LLMs on internal manuals, error logs, and historical maintenance data. - Integrating AI assistant with embedded telemetry modules for real-time diagnostics. - Conducting fault analysis and implementing preventive maintenance strategies.

AI Intern
Tomatiki, Egypt
Dec 2024 – Mar 2025 - Developed ML models for predicting equipment failures. - Used LLMs to generate explanations and suggestions for maintenance actions.

Embedded System Developer (Freelance)
Feb 2024 – Present - Designed and deployed automation prototypes with sensor integration. - Built embedded systems with UART, PWM, and EEPROM logic.

Embedded Systems Intern
Egyptian Engineering Foundation
May 2024 – Aug 2024 - Built and tested embedded control circuits. - Applied communication protocols such as UART, I2C, SPI.

QA Intern
Oigetit Fake News Filter (Remote)
- Executed manual and automated tests using Selenium and TestNG. - Created bug reports and collaborated with international QA teams.

Freelance Instructor
- Delivered basic to intermediate training in embedded systems and programming (C/C++).

Education

Bachelor of Mechatronics Engineering
October 6 University
2018 – 2023
- **GPA:** 3.17 (Very Good)
- **Graduation Project:** ROV for underwater pipeline inspection using Raspberry Pi, camera, IMU & pressure sensors.
- **Grade:** Excellent

Key Projects

- **Proof of Concept (PoC) Project:** GitHub Repository – POC project
  - CNC-focused RAG chatbot trained on internal manuals and maintenance logs to assist technicians and customers. *(In Progress)*
- **Digital Twin Simulation:** Unity-based production line simulation integrated with Arduino telemetry.
- **Arabic OCR Water Meter Reader:** YOLOv11 + EasyOCR model trained on Roboflow dataset.
- **Fridge Fault Prediction:** ANN model to detect cooling failures with >95% accuracy.
- **Smart Parking System:** IR sensors + EEPROM + PWM motor control.
- **Temperature Hazard System:** Real-time alerting system using Atmega328p.

Technical Skills

- **AI & Machine Learning:** TensorFlow, OpenCV, EasyOCR, YOLOv5/11, Huggingface, RAG, Prompt Engineering
- **Programming Languages:** Python, C, C++, Java, C#
- **Embedded Systems:** Arduino, Atmega328p, PIC, UART, I2C, SPI, EEPROM, Proteus
- **Digital Twin & Simulation:** Unity, Omniverse Isaac Sim, OpenUSD
- **Backend Development:** FastAPI
- **Databases:** PostgreSQL, FAISS, Chroma
- **Tools & Platforms:** Git, GitHub, Selenium, TestNG, Jira, Trello
- **Control Systems:** TIA Portal, Ladder Logic, Classic Controls

Certifications & Courses

- NTI Digital Twin Program (2025)
- Robotics & Mechatronics 3: Unity Digital Twin (2025)

- Testing Bootcamp – Tarek Roshdy (2024)
- Classic Control & PLC (2021)
- Advanced C++ Programming – IT Sharks (2022)
- Interfacing with PIC & AVR Microcontrollers

Languages

- Arabic: Native
- English: Very Good

Personal Information

- **Date of Birth:** 19/09/1999
- **Gender:** Male
- **Nationality:** Egyptian
- **Military Status:** Exempted